Tackling Waste Misclassification: Building a Deep Learning System for Smarter Garbage Sorting

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Abstract

Improper waste disposal remains a major environmental issue. One way to remedy this is by creating a garbage classification system. However, such a system is very difficult to generalize to the infinite shapes and sizes of arbitrary objects, and major misclassifications can lead to environmental pollution during recycling. Our project leverages deep learning and computer vision to build a robust garbage classification model that can identify waste categories to the highest degree of accuracy. Specifically, our waste classification system is built on a ResNet-50 backbone pre-trained on ImageNet, enhanced with a Convolutional Block Attention Module (CBAM) to improve feature refinement after the backbone output. Our model achieves near-perfect performance with a Top-5 score of 99.97% and an F1 score of 95.82%.

1 Introduction

We decided to focus on garbage classification because it represents a highly useful real-world application. In the commercial sector, most garbage bins mix all waste due to the overhead and inefficiency of manual sorting. At home, individuals often either lack the judgment or the knowledge required to classify waste items correctly and end up throwing everything in one bin. By creating a garbage classification model, we could help improve recycling efficiency and reduce environmental harm!

Additionally, our model has several useful applications, from as simple as being a mobile app to something more advanced, like being integrated into robot cleaners, smart bins, or other smart household devices, further streamlining eco-friendly waste sorting. Since many of us have younger siblings, we also thought this would be a perfect way for them to learn about sustainability.

The dataset chosen to map this problem was chosen with great precision and foresight. We selected the the highest rated dataset on Kaggle in our domain, the Garbage Dataset (Kunwar, 2024) [2]. This dataset contains roughly 20k pictures of garbage items and contains 10 of the most prominent garbage types as its classes. This dataset includes both controlled (lab-style) and real-world images, providing a good variety and realism for training a machine vision model that will be used to generalize various garbage types in the real-world.

Figure 1: Sample image for each class in dataset

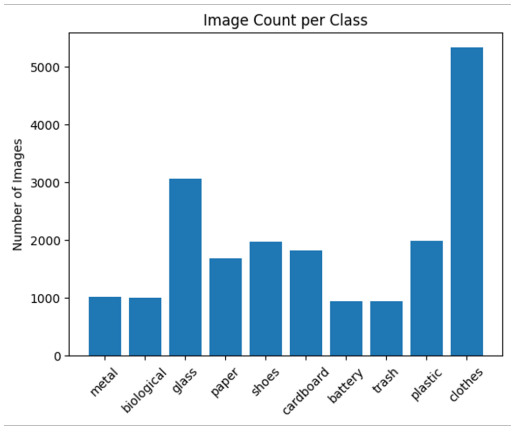

Figure 2: Distribution of classes in dataset

This is a supervised learning classification problem where we have trained a powerful garbage classifier that learns from this meticulously curated labeled data. The input to the system is a 2D image of an object (JPG), and the output is 1 of the 10 garbage classes as seen in Figure 5 and 6 above.

Why we really want to solve this garbage classification problem:

- Working on a problem with direct applications to saving the environment!
- Heavily researched problem, allowing us to utilize transfer learning more effectively to extend novel architectures to map the problem and achieve high performance seamlessly.
- Small enough dataset (less than 1GB), allowing us to create various ensembles, leading to better generalizations to real-world scenarios.
- Opportunity to apply novel architectural changes to deep neural networks (such as attention mechanism with CBAM), enabling our network to learn sharp discriminative features like no other, potentially leading to true generalization.

2 Related Work

When it comes to automated waste classification using machine learning, several prior studies have already laid the groundwork. Some of the ones we read into are:

- Md. Nahiduzzaman (2025) [4] a waste sorting system using an ensemble of ResNet50 and EfficientNet models, integrated into a smart bin for real-world deployment
- Mao (2021) [3] a baseline CNN trained from scratch on common waste types, achieving modest accuracy
- Al-Mamun (2023) [1] an optimized CNN with hyperparameter tuning and model pruning to distinguish visually similar classes efficiently

When it came to designing our system, we first followed Mao's approach by creating two baseline models. We opted for ResNet-50 and Vision Transformer, and similarly achieved modest accuray (see "Epxeriments" section below). From there, we got inspired by how Al-Mamun used different optimization techniques to enhance his model. However, we didn't exactly follow his footprints. Rather we used transfer learning to leverage pre-trained features, convolutional block attention modules to focus on the most informative regions of the images, and multi-scale feature fusion to capture features at multiple resolutions. This was our own twist we added, as it was very effective.

3 Model Architecture

3.1 Figure or diagram

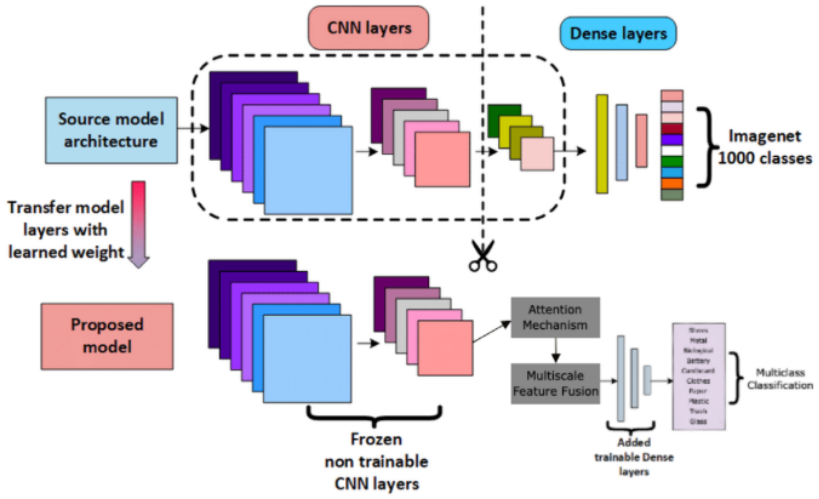

Figure 3: The EcoSort Architecture Pipeline.

This diagram illustrates our waste classification model, which begins by transferring the weights from an existing ResNet-50 backbone. The flow highlights our two key architectural innovations: the CBAM unit, which refines high-level features by attending to the object rather than the background, and the Multi-Scale Feature Fusion (MSFF) module. The MSFF specifically visualizes the skip-connection strategy, where detailed texture information is fused with semantic information prior to the final classification head.

4 Clear Description

4.1 Problem Domain

The core problem EcoSort addresses is the difficulty of classifying waste in real-world environments. Unlike clean, stock photos of objects, real-world waste is often deformed; crushed cans and crumpled paper do not look like their original shapes. Furthermore, waste is often photographed against messy backgrounds (floors, bins, sidewalks), creating noise that confuses standard classifiers. We utilize the Garbage Classification V2 dataset, which contains 10 distinct classes, to train and evaluate our solution.

4.2 Model Architecture

To address the complexity of visual waste identification, we designed a custom architecture based on ResNet-50. Our model improves upon the standard backbone by integrating attention mechanisms and feature fusion to capture both high-level semantic meaning and low-level textural details.

The architecture consists of four sequential stages:

- **Backbone (ResNet-50):** We utilize the convolutional layers of ResNet-50, pre-trained on ImageNet. This allows the model to immediately recognize basic shapes and edges without training from scratch.
- **Attention Mechanism (CBAM):** We insert a Convolutional Block Attention Module (CBAM) after the final feature layer. This block acts as a filter that highlights relevant features (the waste object) and suppresses irrelevant ones (the background noise).

- **Multi-Scale Feature Fusion (MSFF):** A distinct innovation in our architecture is the MSFF module. Standard CNNs often lose fine-grained details in deep layers. Our MSFF module fuses "Deep" semantic features with "Shallow" textural features via element-wise summation. This ensures the model considers both the object's shape and its surface texture (e.g., distinguishing metal from plastic).
- **Classification Head:** The final stage is a fully connected layer with Dropout ($p = 0.3$) to map the fused features to the 10 waste categories.

4.3 Methodology and Implementation

4.3.1 Data Preprocessing and Augmentation

Before entering the network, images undergo a transformation pipeline. All inputs are resized to 256×256 and normalized using ImageNet statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$).

To prevent overfitting on the limited dataset, we apply aggressive augmentation during training:

- Random Resized Crop: Scales images (0.8–1.0) to force the model to recognize objects at different zoom levels.
- Geometric Transformations: RandomHorizontalFlip ($p = 0.5$) and RandomRotation ($30°$) to handle orientation variance.
- Photometric Distortions: ColorJitter (brightness, contrast, saturation=0.3, hue=0.1) to simulate varying lighting conditions.

4.3.2 Optimization Strategy

We utilize the Adam optimizer with L_2 regularization (weight decay = $1e^{-4}$) to smooth the loss landscape. To ensure convergence, we implemented a `ReduceLROnPlateau` scheduler. If the validation loss stagnates for 3 epochs, the scheduler automatically halves the learning rate, allowing the model to make finer adjustments to the weights.

4.3.3 Two-Phase Training Algorithm

To adapt the pre-trained weights to our specific domain, we implemented a distinct two-phase training loop:

1. **Phase 1: Warm-Up (5 Epochs):** We freeze all parameters in the ResNet-50 backbone (`requires_grad = False`). We train only the custom modules (CBAM, MSFF, and Classifier) with a learning rate of $1e^{-3}$. This allows the custom layers to align with the backbone features before we alter the backbone itself.
2. **Phase 2: Fine-Tuning (15 Epochs):** We unfreeze the deep layers of the backbone. We then retrain the entire network with a significantly lower learning rate ($1e^{-4}$). This allows the feature extractors to adapt specifically to waste textures (e.g., differentiating crushed aluminum from paper) while retaining the robust edge detection learned from ImageNet. We implemented an early stopping mechanism with a patience of 5 epochs. If the validation accuracy plateaus or stops improving in a meaningful way, the training is stopped to prevent overfitting.

4.4 Loss Function

We optimize the model using the Cross-Entropy Loss function. Mathematically, it calculates a penalty based on the divergence between the predicted probability distribution and the actual class label:

$$\mathcal{L} = -\sum_{c=1}^{M} y_{o,c} \log(p_{o,c})$$

Where:

- $\mathcal{L}$ is the loss.

- M is the number of waste classes (10).
- $y_{o,c}$ is the binary indicator (0 or 1) if class c is the correct classification.
- $p_{o,c}$ is the predicted probability that observation o belongs to class c.

5 Comparison or demonstration / Experiments

The overall goal of our project was to develop an automated image classification system capable of accurately identifying different types of garbage. So, in order to meet that goal, we used multiple metrics including Top-1 and Top-5 accuracy, F1 score, and confusion matrices. This allowed us to not only assess the overall and per-class performance of any model, but is also allowed us to identify any common misclassification patterns. Before training, we augmented some images from our dataset by applying random transformations such as flipping and rotating them. The reason we did this was to better simulate real-world variations in garbage as not only images of garbage will be in good quality.

Model	Top-1	Top-5	F1 (Macro)
ResNet-50	70.49%	96.49%	61.88%
Vision Transformer	75.30%	97.20%	67.50%
ResNet-50 (Transfer Learning)	96.09%	99.83%	95.15%
ResNet-50 (Transfer Learning, CBAM, Multi Feature Fusion)	**96.56%**	**99.97%**	**95.82%**

Figure 4: Quantitative comparison across trained models

From our experiments, we saw that our baseline models (ResNet-50 and ViT), which we trained from scratch, performed the weakest. They achieved 70-75% top-1 accuracy and 61-67% F1-score. We were able to conclude our baseline models were overfitting as they were not able to generalize beyond the training images. However, our ResNet-50 with transfer learning showed a significant improvement, achieving 96% top-1 accuracy and 95% F1-score. This showed that transfer learning allowed the model to leverage pre-trained features, enabling it to generalize much better to unseen data. Lastly, with a convolutional block attention module (CBAM) and multi-scale feature fusion (MSFF) on top of the ResNet-50 with transfer learning model, we again saw a slight improvement in performance. This version achieved 96.5% top-1 accuracy and 96% F1-score, making it the strongest model in our evaluation.

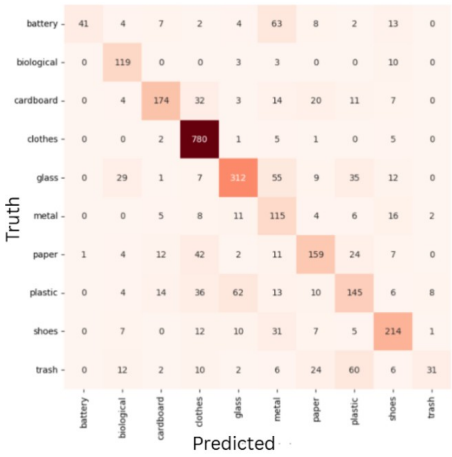

Figure 5: Qualitative result of worst model

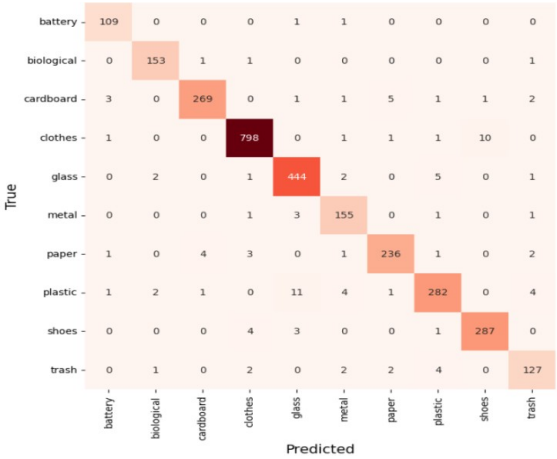

Figure 6: Qualitative result of best model

From the confusion matrix of the baseline ResNet-50 model, as shown in Figure 5 above, we observed a high degree of confusion between garbage made from visually similar materials. Even after multiple

training epochs, we noticed that the model confused plastic with glass, batteries with metal, and trash with plastic. In contrast, the ResNet-50 model with transfer learning, CBAM, and MSFF avoided these errors for the most part, as shown in figure 6 above. This meant the model it was less confused as it was able to differentiate between garbage made from similar looking materials more effectively.

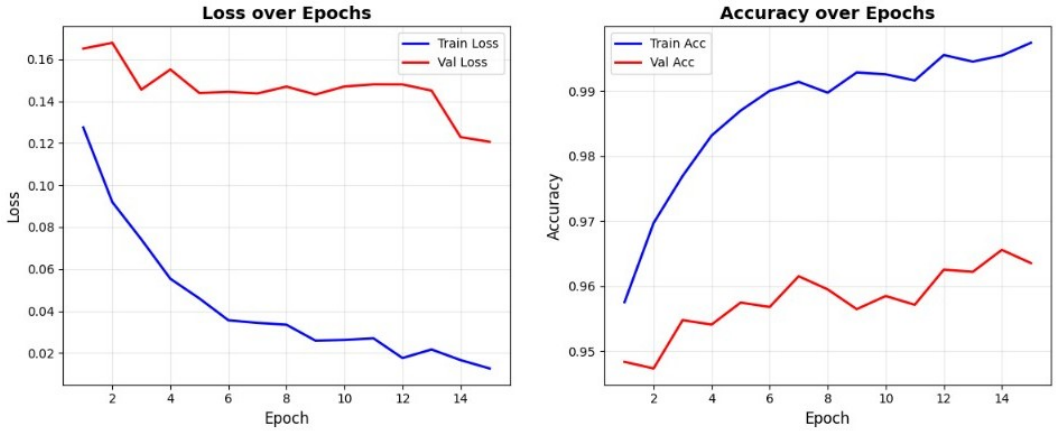

Figure 7: Shows training and validation error of best model

In the image above, the left graph shows the loss over multiple epochs. As epochs increased, the training loss steadily decreased and the validation loss remained pretty much stable. This showed that our model was effectively learning without overfitting. The graph on the right shows accuracy over the multiple epochs. As epochs increased, both the training and validation accuracy increased, demonstrating strong generalization.

6 More Into Our Novelty

The novel aspects of this model were derived with an intention to improve upon an existing architecture, designed and built upon ResNet-50, for the high-noise, texture-reliant domain of waste classification. Initially, we noticed that our baseline ResNet-50 and ViT models were not adequate in picking up discriminative features for different materials (as indicated from our results above), hence we had to extend some feature(s) that could overcome this issue. Fortunately, in CSC413, the topic of the attention mechanism was brought up, and it conceptually aligned with our model and the discrimination issue we were looking to resolve. Therefore, this became our primary motivation for integrating attention modules (and this turned out to be a very fruitful prediction)! Upon doing additional research, it was discovered that a feature fusion mechanism could also bring more detail to the model in order to better pick up discriminative features and object location within a given image.

Hence, the final model's approach injects two distinct mechanisms, an attention mechanism, and a feature fusion mechanism to address the specific shortcomings of standard CNNs in less structured environments, where things like background noise can significantly impact the classification.

The first of these aspects was an attention mechanism, in particular, the Convolutional Block Attention Module (CBAM). This module allowed for refining of the feature map after the frozen layers present in the ResNet. It allows our model to better think about the importance of features in the map. This is accomplished through pooling average and maximum values, and attention modules.

1. **Channel Attention Module:** By utilizing both average and maximum pooling, the model calculates which feature channels are most significant (e.g., emphasizing "metallic reflection" over "grass texture").

2. **Spatial Attention Module:** This step generates a spatial mask that again aims to highlight the location of the object in the image, ignoring background noise.

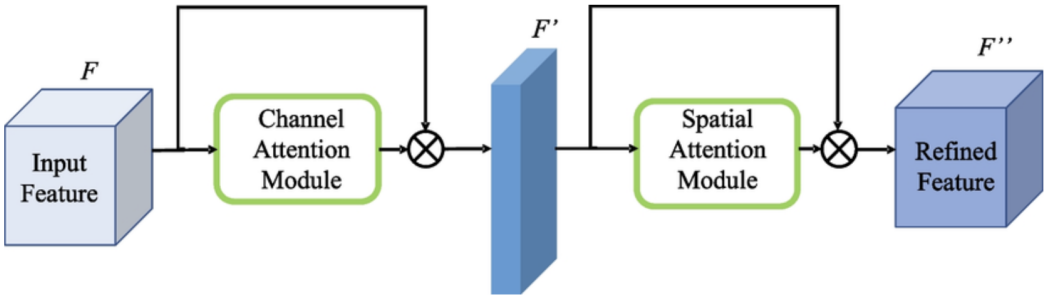

Figure 8: CBAM diagram

The second of these aspects was a Multi-Scale Feature Fusion (MSFF) which allowed the model to extract feature maps from the existing layers and fuse them with output map of the CBAM. This was implemented, as a limitation that was identified in the initial model was the loss of finer grained textural information. This loss occurred due to the immense amount of processing that eventually left the output map with features more focused on broad shape.

Both of these aspects allowed for us to see statistically significant improvements in the model's accuracy, particularly in distinguishing classes with high visual similarity. When testing, the addition of both of these mechanisms, it was critical to take a look at what images were correctly classified by the updated model in comparison to the ones which were incorrect on the older model. This presented confirmation that the updated model performed better on images with high similarity, and minor distinctions, while also demonstrating the elimination of background noise which plagued some classifications. In addition, a simplified classification head also improved accuracy as the feature map produced by the new model was significantly more informative.

7 Limitations

7.1 Visual Similarity of Materials

From a technical standpoint, garbage comes in all shapes/sizes/materials, and the images all vary in lighting/angle/colour. As a result, items such as glass and clear plastic can look nearly identical.

Figure 9: Plastic bottle in comparison to a glass bottle

From the human eye, the distinction between plastic and glass is apparent as we know the common shapes and textures from experience, enabling us to seamlessly distinguish in the 3D space. However, from the eye of the computer (the computer vision model), extracting discriminative features from 2D images of such visually similar objects is inherently challenging. Our final model was powerful enough to discern such objects from our training, validation and tests sets, however, with a certain degree of added lighting, angle or colour, the model would periodically fail (misclassify) for these materials with visual similarity.

A possible extension to address this limitation would be to spend more time on data augmentation. Since we noted above that the model may fail in very specific scenarios with an added degree of lighting/angle/colour, this suggests focusing heavily on generalization to quirky/erratic images that are very unexpected. Further data augmentation that could have been performed to tackle this issue can include random cutout/erasing of small portions of the images, or some gaussian noise injection for the images.

7.2 Dataset Imbalance and Minority Class Performance

A significant limitation identified during our data analysis was the severe class imbalance within the *Garbage Classification V2* dataset. As noted in our distribution plots, classes such as "Clothes" contained over 5,000 samples, whereas "Battery" and "Trash" contained fewer than 1,000 images.

While we employed a stratified train-test split to ensure proportional representation, the model still had significantly fewer opportunities to learn the feature representations of the minority classes. Consequently, the model exhibits a bias towards the majority classes. In scenarios where a battery is photographed in a way that slightly resembles a texture found in clothing or paper, the model is statistically more likely to predict the majority class simply because it minimizes the global loss more effectively during training.

A future extension to mitigate this would be the implementation of a Weighted Cross-Entropy Loss. By assigning higher penalty weights to misclassifications of minority classes (e.g., penalizing a wrong "Battery" prediction 5x more than a wrong "Clothes" prediction), we could force the optimizer to prioritize learning the rarer items. Alternatively, synthetic data generation (using GANs) could be used to artificially inflate the size of the minority classes.

8 Conclusions

We successfully developed a garbage classification system using deep learning and computer vision, achieving our goal of creating a model designed for real-world impact. By assessing multiple different models throughout our implementation phase, we found that leveraging transfer learning and attention mechanisms for classification tasks provided the best accuray. Our final model demonstrated strong generalization and reliability by accurately classifying waste with a high degree of precision and confidence.

9 Individual Contributions

The development of the EcoSort project was a collaborative effort, with each team member owning specific technical components of the pipeline while maintaining close synchronization on the overall architecture.

- **Dev:** Responsible for the foundational research of existing literature and establishing the project structure. He implemented the data pre-processing pipeline, including augmentation strategies, and set up the collaborative Jupyter notebook environment.
- **Ishan:** Focused on establishing a performance benchmark. He built and trained the baseline model using standard Transfer Learning on ResNet-50. This baseline was critical for validating the effectiveness of our proposed novel architecture by providing a point of comparison.
- **Om:** Conducted initial research alongside Dev and investigated potential areas for architectural novelty. He was responsible for the detailed analysis of evaluation results (e.g., confusion matrices and per-class accuracy), providing the insights necessary to fine-tune the model.
- **Ishaan:** Led the research and implementation of the project's novel contributions. He coded the Convolutional Block Attention Module (CBAM) and the Multi-Scale Feature Fusion (MSFF) module, integrating them into the training loop to create the final EcoSort architecture.

Coordination: The team held weekly meetings via Discord to synchronize progress, debug implementation issues, and collaboratively write the final report.

References

[1] Al-Mamun. Recycling waste classification using an optimized convolutional neural network. 2023.

[2] S. Kunwar. Garbage dataset, 2024.

[3] Mao. Garbage classification using convolutional neural networks. 2021.

[4] M. Nahiduzzaman. An automated waste classification system using deep learning. 2025.